AS · Cambridge (CIE) · Physics

20 mins 20 questions

Multiple Choice Questions

Progressive Waves

Progressive Waves / Cathode-Ray Oscilloscope / The Wave Equation / Wave Intensity

Easy (7 questions)	/7
Medium (6 questions)	/6
Hard (7 questions)	/7
Total Marks	**/20**

Scan here to return to the course
or visit savemyexams.com

© 2026 Save My Exams, Ltd.

Easy Questions

1 A wave has a frequency of 4 GHz.

What is the period of the wave?

A. 25 000 μs

B. 250 ps

C. 25 ns

D. 2.5 ns

(1 mark)

2 A light with wavelength of 600 nm has an orange colour.

What is the frequency of this light?

A. 500 kHz

B. 500 THz

C. 180 GHz

D. 180 Hz

(1 mark)

3 A sound wave has a frequency of 50 Hz and a speed of 330 m s^{-1}

If two waves have a phase difference of 60°, what is the possible distance between two points on the wave?

A. 6.6 m

B. 2.2 m

C. 1.1 m

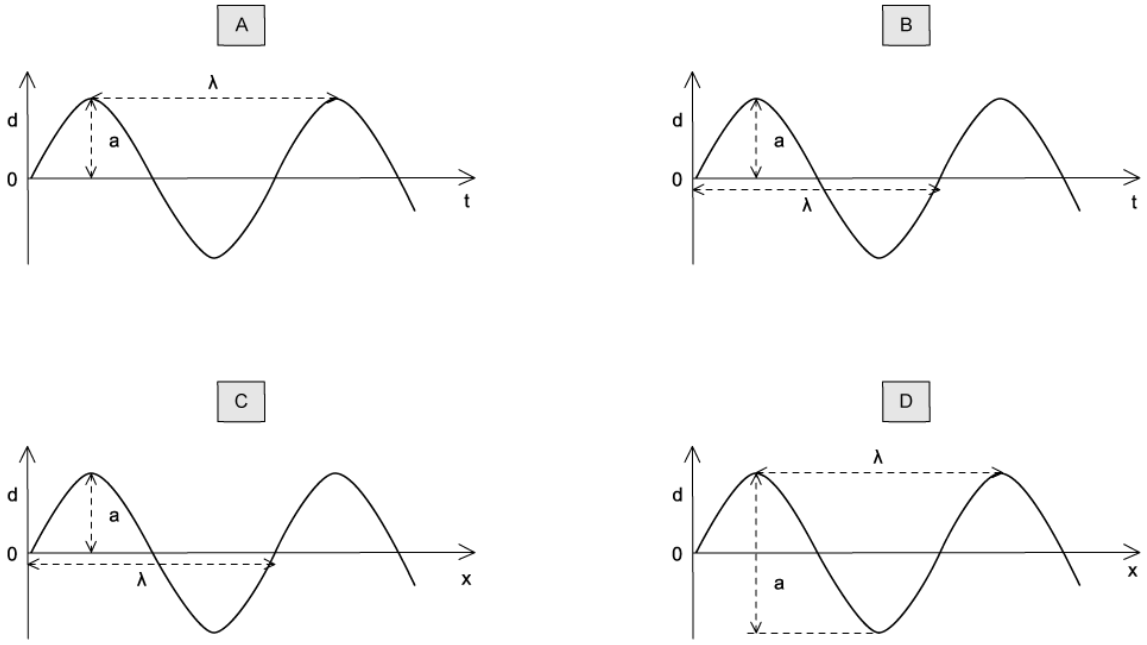
D. 0.03 m

(1 mark)

4 A sound wave has a displacement *y* at a distance *x* from its source at time *t*.

© 2026 Save My Exams, Ltd.

Which graph shows correctly the amplitude a and the wavelength λ of the wave ?



(1 mark)

5 The following are statements about waves.

1	Longitudinal waves can be polarised.
2	The amplitude of a wave is directly proportional to the energy transferred by the wave.
3	All electromagnetic waves travel at the same speed in a vacuum.
4	The frequency of infra-red light is greater than the frequency of ultra-violet light.

Which ones are incorrect?

A. 1, 3 and 4

B. 1, 2 and 4

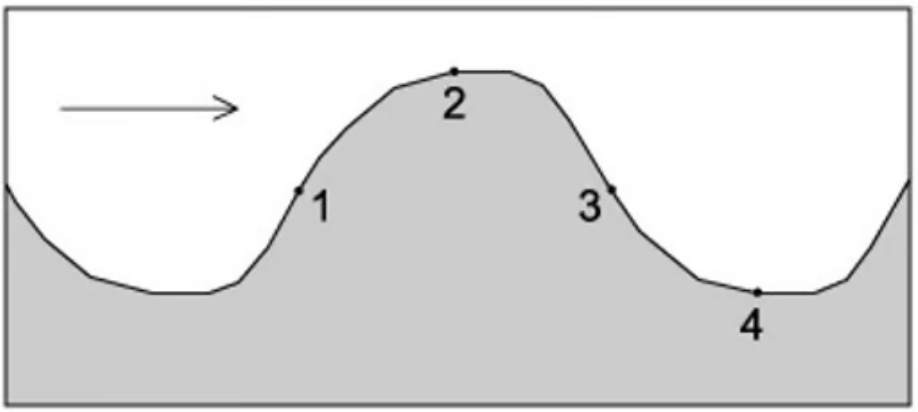
© 2026 Save My Exams, Ltd.

C. 1, 2 and 3

D. 2, 3 and 4

(1 mark)

6 The vertical cross-section diagram shows a water wave moving from left to right.



Which points on the diagram show the water moving upwards with maximum speed?

A. 1 and 3

B. 3 only

C. 1 and 4

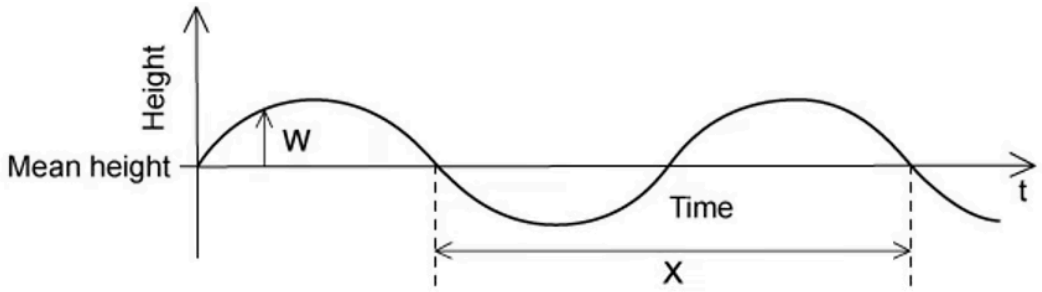
D. 4 only

(1 mark)

7 The diagram below is a graph showing how the height of water at a point in a marina varies

© 2026 Save My Exams, Ltd.

with time *t* as the waves pass the point.



What is the correct label for W and X?

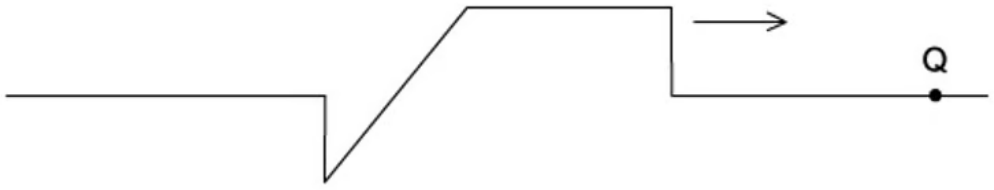
	W	X
A	amplitude	period
B	displacement	wavelength
C	amplitude	wavelength
D	displacement	period

(1 mark)

© 2026 Save My Exams, Ltd.

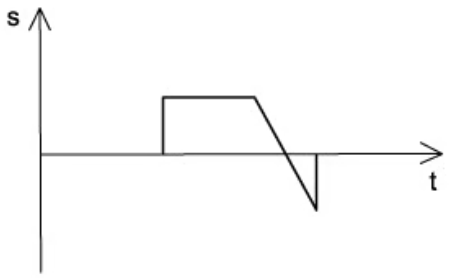
Medium Questions

1 The diagram shows a wave moving along a stretched rope in the direction shown.

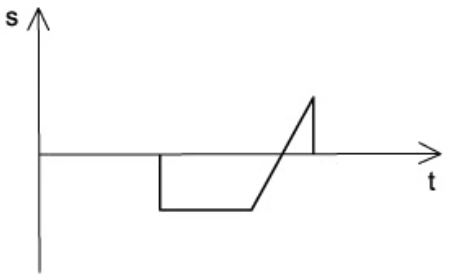


Which of the following will be the correct graph to show the variation with time *t* of the displacement *s* of the particle Q in the rope ?

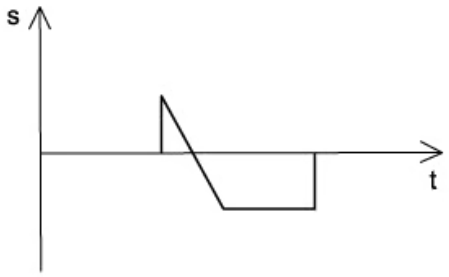
A



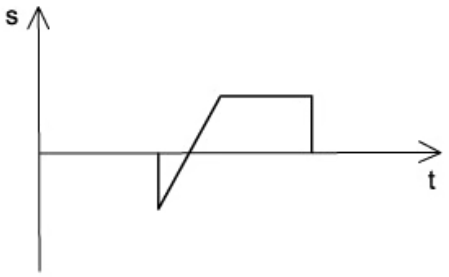
B



C



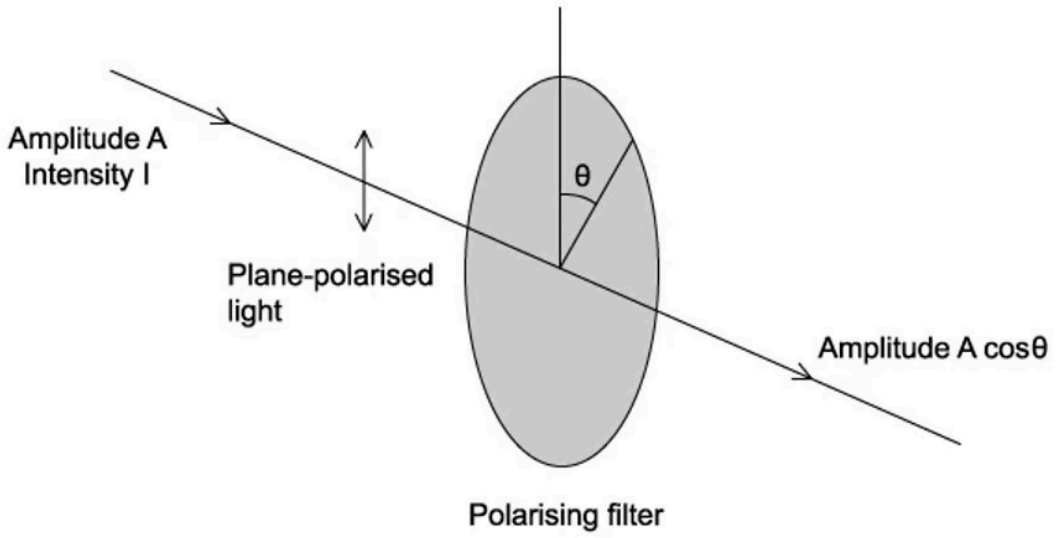
D



(1 mark)

© 2026 Save My Exams, Ltd.

2 A plane polarised light of amplitude *A* is passed through a polarising filter as show in the diagram. The emerging light has an amplitude of A cos θ



If the intensity of the beam is *I*, what is the intensity of the emerging light when θ is 60.0°?

A. 0.250 *I*

B. 0.500 *I*

C. 0.750 *I*

D. 0.866 *I*

(1 mark)

3 A sound wave with an amplitude *a* of 0.2 mm has an intensity of 2.5 W m^{-2}

If a wave with the same frequency has an amplitude of 2*a* what is the intensity of the wave?

A. 10.0 W m^{-2}

B. 9.0 W m^{-2}

C. 6.0 W m^{-2}

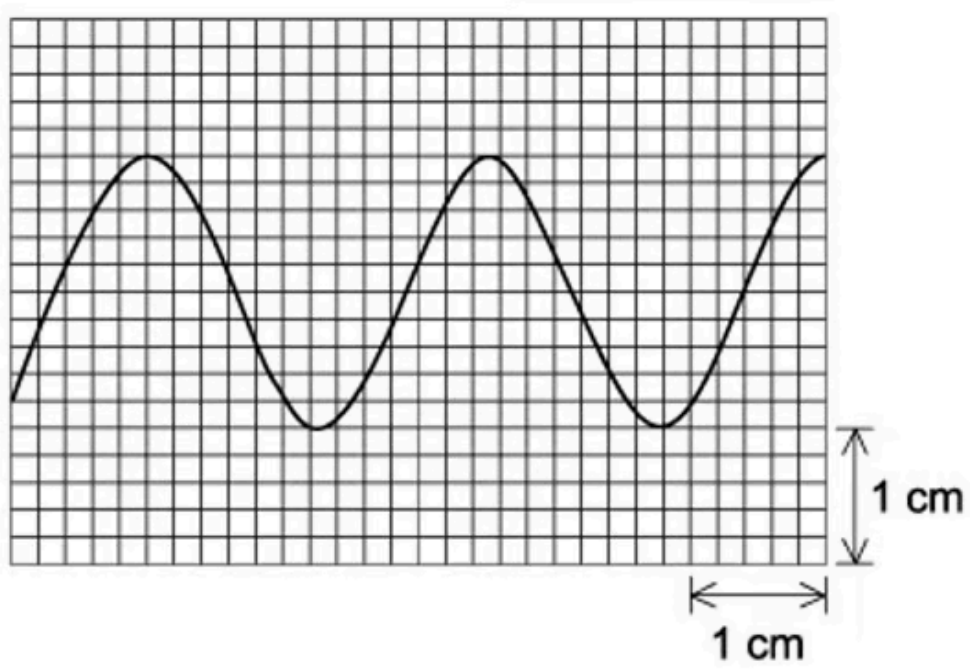
D. 4.2 W m^{-2}

(1 mark)

4 A sound wave can be displayed on a cathode ray oscilloscope (C.R.O.). The time base is set to

© 2026 Save My Exams, Ltd.

5 μs mm^{-1}.



What is the frequency of the sound wave?

A. 5 Hz

B. 100 Hz

C. 5 kHz

D. 100 kHz

(1 mark)

5 A source of sound power P is situated in an open space. The intensity I of sound at distance r from this source is given by

$$I = \frac{P}{4\pi r^2}$$

How does the amplitude A of the vibrating air molecules vary with the distance r from the source ?

A. $A \propto \frac{1}{r}$

© 2026 Save My Exams, Ltd.

B. $A \propto \frac{1}{r^2}$

C. $A \propto r$

D. $A \propto r^2$

(1 mark)

6 Radio waves are electromagnetic waves that travel at speed c in a vacuum with a frequency f and wavelength λ.

Which row in the table describes the wavelength and speed of radio waves with a frequency $3f$?

	Speed in a vacuum	Wavelength
A	$3c$	3λ
B	c	$\frac{\lambda}{3}$
C	c	3λ
D	$\frac{c}{3}$	$\frac{\lambda}{3}$

(1 mark)

© 2026 Save My Exams, Ltd.

Hard Questions

1 The equation below shows the speed of *v* waves in deep water

$$v^2 = \frac{g\lambda}{2\pi}$$

Where *g* is the acceleration of free fall and λ is the wavelength of the waves. The wavelength λ and the frequency *f* of the wave were measured.

If the graph was to give a straight line through the origin which of the following graphs should be plotted?

A. *f* against λ^2

B. f^2 against λ

C. *f* against $\frac{1}{\lambda}$

D. f^2 against $\frac{1}{\lambda}$

(1 mark)

2 A light wave is incident normally on a surface of area *S* with an amplitude *A*. The power per unit area reaching the surface is *P*.

If the light is focused on a smaller area of $\frac{1}{3}S$, and the amplitude of wave is doubled to 2*A* what is the power per unit area?

A. 6*P*

B. 12*P*

C. 18*P*

D. 36*P*

(1 mark)

3 All electromagnetic waves travel at speed *c* in a vacuum with a frequency *f* and wavelength λ.

© 2026 Save My Exams, Ltd.

Which row in the table describes the wavelength and speed of electromagnetic waves with a frequency $\frac{f}{2}$?

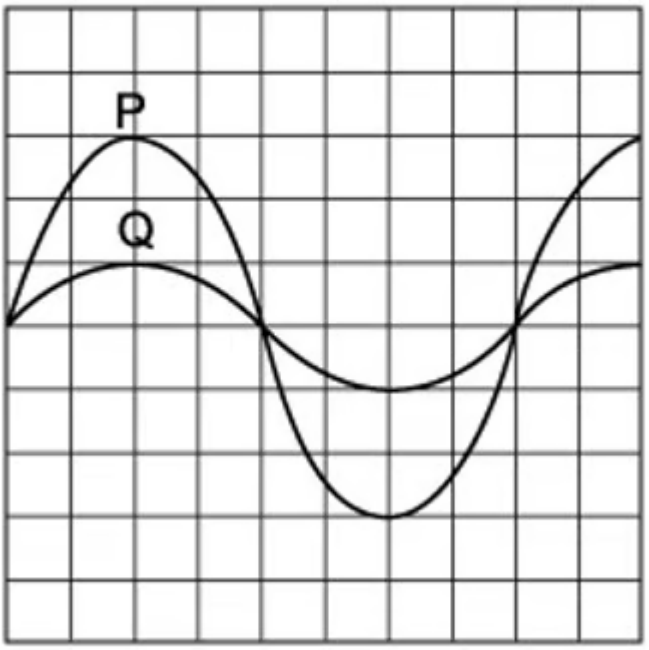
	Speed in a vacuum	Wavelength
A	$2c$	2λ
B	c	2λ
C	c	$\frac{\lambda}{2}$
D	$\frac{c}{2}$	$\frac{\lambda}{2}$

(1 mark)

4 Sound waves can be displayed on a cathode ray oscilloscope (C.R.O.). The graph below represents two waves seen on the screen. They are labelled P and Q as shown in the

© 2026 Save My Exams, Ltd.

diagram



What is the ratio $\frac{intensity\ of\ sound\ wave\ P}{intensity\ of\ sound\ wave\ Q}$?

A. $\frac{\sqrt{3}}{1}$

B. $\frac{9}{1}$

C. $\frac{1}{1}$

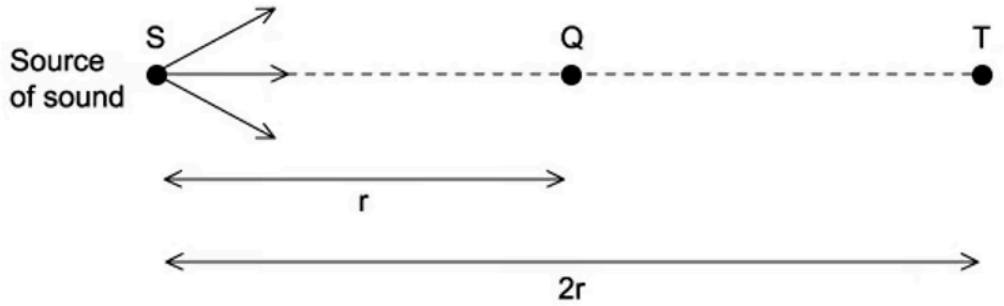
D. $\frac{3}{1}$

(1 mark)

5 The intensity *I* of a sound at point Q is inversely proportional to the square of the distance *x* of Q from the source of the sound. This can be shown by the equation

© 2026 Save My Exams, Ltd.

$$I \propto \frac{1}{x^2}$$



Molecules of air at Q are a distance *r* from S and oscillate with amplitude 8.0 μm.

Point T is a distance 2*r* from S.

What is the amplitude of oscillation of air molecules at T?

A. 4.0 μm

B. 2.0 μm

C. 2.8 μm

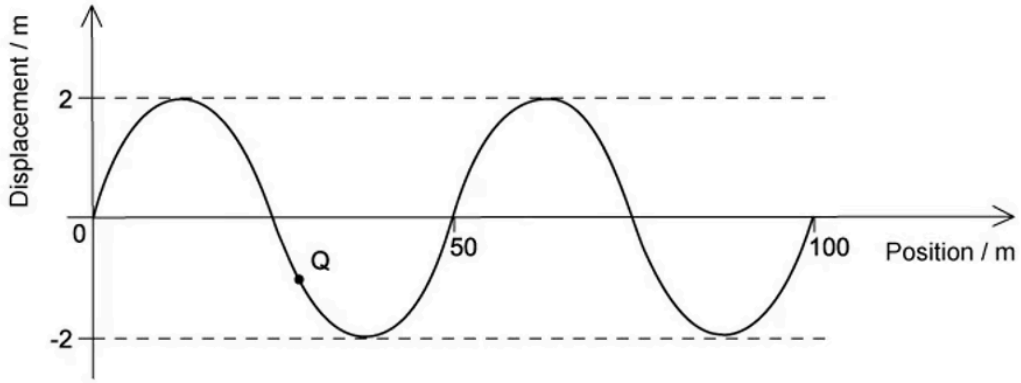
D. 1.4 μm

(1 mark)

6 The graph shows a sinusoidal wave in water travelling at a speed of 8.0 m s^{-1}.

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com

The maximum speed of the particles in the wave is $2\pi af$, where a is the amplitude and f is the frequency.



An object Q is floating on the water it has a mass of 2.5×10^{-3} kg.

What is the maximum kinetic energy of Q due to the wave? Assume the motion is vertical.

A. 4.1 mJ

B. 5.1 mJ

C. 39 mJ

D. 64 mJ

(1 mark)

7 A wave has a speed of 340 m s^{-1} and a frequency of 500 Hz

If two points on a wave are 0.17 m apart, what is the phase difference?

A. $\frac{\pi}{4}$ rad

B. $\frac{\pi}{2}$ rad

C. $\frac{3\pi}{4}$ rad

D. π rad

(1 mark)

© 2026 Save My Exams, Ltd. | Get more and ace your exams at savemyexams.com